

Literature review

Peroneal reflex contribution to the development of functional instability of the ankle joint

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Abstract

Sprain of the lateral ligament complex of the ankle joint is one of the most common athletic injuries. Following initial injury a large percentage of individuals develop complaints of repetitive ankle sprain, such that they feel that the ankle “gives way” or feels unstable. These complaints of “giving way” and the occurrence of repeated inversion injury have been termed functional instability of the ankle joint.

Previously it was proposed that functional instability developed as a result of articular deafferentation, whereby damage to mechanoreceptors in the ankle joint supporting ligaments and capsule at the time of initial injury resulted in reduced afferent feedback to the evertor musculature of the ankle joint. Consequently it was hypothesized that the evertor musculature could not reflexively stabilize the ankle joint if it was subjected to a sudden inversion perturbation. The aim of this paper was to review the literature relating to the reflex response times of the evertor musculature of the ankle joint to sudden unexpected inversion perturbation, and thus, discuss the role of reflex activity in the peroneal muscles to the development of functional instability of the ankle joint. There is conflicting evidence regarding the extent to which ligament mechanoreceptors induce reflex stabilization of the ankle joint in response to rapid inversion movements. Even if such reflex mechanisms do exist it is unlikely that they would be fast enough to provide adequate joint protection during dynamic activity, suggesting that their actual functional relevance is limited.

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1. Introduction

Ankle injuries commonly occur in sports with a high level of jumping and cutting activities, and especially in ball sports (Ekstrand & Tropp, 1990) and may account for up to 25% of all time lost from competition (Mack, 1982). The most prevalent mechanism of injury is one of inversion, plantar flexion, and internal rotation (Garrick, 1977), a position in which the lateral ligament complex and in particular the anterior talofibular ligament is subject to the greatest amount of strain (Colville, Marder, Boyle, & Zarins, 1990). Garrick (1977) indicated that injury to the lateral ligament

complex may typically occur in two ways. The first of these involves a poorly executed cutting manoeuvre which can be enhanced by inadvertent foot fixation as with long, conical football cleats and studs on grass. The second mechanism involves alighting on an irregular playing surface (e.g. a rutted playing field or another player's foot).

Ankle sprains usually occur in one of three categories: lateral, medial, or syndesmosis. Eighty-five percent of all ankle sprains are lateral sprains, eversion sprains comprise 5% of all sprains, and syndesmosis sprains comprise 10% of these injuries (Baldwin & Tetzlaff, 1982). Even though significant disability is not a common sequel of most ankle sprains, residual pain and functional problems following lateral ligament complex sprains are often reported in the literature

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(Bosien, Staples, & Russell, 1955; Brand, Black, & Cox, 1977; Freeman, Dean, & Hanham, 1965; Gerber, Williams, Scoville, Arciero, & Taylor, 1998; Löfvenberg, Kärrholm, & Lund, 1994).

In a retrospective study on ankle sprains, Staples (1972) reported that less than 3% of his patients were seriously disabled; however, more than 40% of patients still had chronic complaints.

Inconsistencies exist in the literature regarding the exact prevalence of chronic ankle dysfunction. Freeman et al. (1965) indicated that chronic ankle dysfunction follows about 40% of injuries of the lateral ligament complex. Ryan (1994) cited three studies which reported that chronic ankle dysfunction complicates 10–20% of all lateral ligament complex sprains presenting for treatment (Moller-Larsen, Wethelund Jurik, Carvalho, & Lucht, 1988; Niedermann et al., 1981; Prins, 1978). Brand et al. (1977) reported the incidence of chronic ankle dysfunction to be 17.5% in a random survey of 175 US Naval Academy athletes.

Typical complaints of patients with chronic ankle dysfunction include pain during activity, recurrent swelling, a feeling of “giving way”, and/or repetitive reinjury (Gerber et al., 1998). The feeling of “giving way” is described as functional instability (Freeman et al., 1965). As this is an often misunderstood area, the aim of this paper is to review reflex contribution of the peroneal muscles to the development of functional instability of the ankle joint.

2. What is functional instability of the ankle joint?

Functional instability of the ankle has been reported to be the most common and serious residual disability following ankle sprains (Karlsson & Lansinger, 1992).

The term functional instability of the ankle joint was first described by Freeman et al. (1965) to classify patients with recurrent complaints of “giving way” of the ankle joint following an initial ankle sprain. Functional instability is a distinctly different phenomenon to that of mechanical instability. Tropp (1985) endeavoured to differentiate functional instability from mechanical instability by defining functional instability as joint motion that does not necessarily exceed an individual's normal physiological range of motion, but which is beyond volitional control; while defining mechanical instability as joint motion that exceeds normal physiological limits. Presently no universally accepted definition exists to classify functional instability of the ankle. Disagreement exists regarding how often “giving way” episodes need to be sustained, how long the disability has had to be present, and at what level of activity injuries are sustained (Konradsen, 2002). Thus, it is essential that in the future a consensus statement is established to develop criteria to help

identify those subjects with functional instability of the ankle.

3. The development of functional instability of the ankle joint

In a recent review, Sjölander, Johansson, and Djupsjöbacka (2002) indicated that it was previously believed that the fundamental role of joint afferents was to prevent injurious ranges of motion by activating protective muscle reflexes. It was hypothesized that when a joint approaches the limit of its normal physiological range of motion that high threshold mechanoreceptors in the supporting ligaments and joint capsule would activate reflex pathways to the muscles surrounding the joint, thus preventing damage due to hyperrotations. Information gathered by the somatosensory, vestibular, and visual systems is processed at three distinct levels of motor control, namely the spinal level, the brain stem level and the higher brain centre level. It has been suggested that the spinal level allows for dynamic muscular stabilization and synchronization of muscle activation patterns based upon afferent input from mechanoreceptors as well as activity from the brain stem and higher brain centre levels (Lephart, Pincivero, & Rozzi, 1998). The latency of muscular activity to involuntary perturbations is used to assess the efficiency of the spinal reflex pathway (Lephart et al., 1998). The role of this neuromuscular pathway during sudden unexpected movements has been evaluated by electromyographic analysis of muscle response times to involuntary perturbations. A deficiency of the muscle activation in response to a sudden unexpected movement could compromise the joint stability.

Freeman et al. (1965) proposed that the basic mechanism of ankle instability following ankle injury develops due to the lesion of mechanoreceptors in the joint capsule and ligaments surrounding the ankle. This proposed theory is known as articular deafferentation. According to this theory, dynamic stability of the ankle joint is dependent upon the ability of the evertors (peronei) to react quickly to sudden inversion perturbations, to develop sufficient tension to prevent injurious ranges of ankle motion, and thus prevent sprains of the lateral ligament complex of the ankle. The theory suggests that individuals with functional instability could have delayed and diminished reflex responses in the evertor muscles of their affected ankle in reaction to an inversion stress due to altered capsular and ligamentous afferent input. It has been suggested that an increase in the response time of the peronei to sudden inversion may have highly significant consequences in terms of risk of injury to the lateral ligament complex of the ankle (Wilkerson & Nitz, 1994). Several studies have investigated this hypothesis, with the main focus of the

studies being the analysis of the reaction of the peronei to sudden unexpected inversion stresses. These studies are outlined in the next section.

4. Studies investigating the response time of the peroneal muscles to sudden inversion perturbation

Konradsen and Ravn (1990) were one of the first authors to investigate the response time of the peroneal muscles to a sudden inversion perturbation of 30°. They measured the surface EMG activity in the peroneus longus and brevis muscles in a group of subjects with functionally unstable ankles. They chose to quantify the time from the moment the trapdoor began to invert, to the first muscle response recorded in the peroneal muscle as the peripheral reflex response of the peronei to the inverting stimulus. Results of the study indicated that the subjects with functionally unstable ankles had a prolonged peripheral reflex response in the peroneus longus and brevis muscles (82 and 84 ms, respectively) when compared to a healthy non-injured group (65 and 69 ms, respectively). Thus the authors concluded that the increase in the peroneal reaction time indicated a deficit in the peripheral reflex stabilization of the ankle joint, consistent with the articular deafferentation theory proposed by Freeman et al. (1965).

Karlsson and Andreasson (1992) comparing the reflex response time (time difference between the rotation of the trap door and the first muscular response registered) of the peronei to sudden inversion perturbation of 30° found that the involved limbs of individuals with unilateral chronic ankle instability demonstrated significantly longer peroneus longus (84.5 ± 4.0 ms versus 68.8 ± 4.5 ms) and peroneus brevis (81.6 ± 5.2 ms versus 69.2 ± 4.1 ms) reflex response times than the uninvolved healthy contralateral limb, thus supporting the results of Konradsen and Ravn (1990).

Löfvenberg, Kärrholm, Sundelin, and Ahlgren (1995) recorded the surface EMG activity of the peroneus longus and tibialis anterior muscles during a sudden unexpected inversion in a group of subjects with chronic ankle instability. In this study the trapdoor was only released when the EMG activity of the muscles under investigation showed a baseline level of activity, thus indicating that the muscles were quiescent and no anticipatory EMG activity was present. The results of this study indicated that the reflex response time of the peroneus longus was significantly longer in the chronic lateral instability group (65 ms) when compared to a control group (49 ms) with no history of chronic lateral ankle instability. This study thus supports the results obtained by previous investigators.

A study by Ebig, Lephart, Burdett, Miller and Pincivero (1997) refuted the hypothesis of articular deafferentation. The purpose of this study was to

investigate the reflex response times of the peroneus longus and tibialis anterior muscles to a sudden plantarflexion/inversion stress in a group with chronic ankle instability—subjects acted as their own controls (i.e. testing was carried out on the self-reported unstable ankle and the previously uninjured contralateral ankle). The reflex response time was quantified as the latency time between the release of the plantarflexing/inverting platform and the onset of EMG activity in the muscles under investigation. The results of the study failed to show a difference in the reflex response time of the peroneus longus (58.6 ± 11.0 ms versus 65.3 ± 17.0 ms) or tibialis anterior (67.9 ± 14.0 ms versus 71.6 ± 14.0 ms) muscles to the sudden unexpected inversion.

Fernandes, Allison, and Hopper (2000) conducted a study to determine whether there was a difference in peroneus longus muscle reflex response time at varying degrees of frontal plane movement in subjects with functional instability compared to a non-injured control group. These authors recorded the reflex response time of the peroneus longus muscle to varying degrees of frontal plane inversion (5°, 10°, and 15°). Results of this study indicated that there was no statistical evidence that history of ankle injury or the subjective complaint of functional instability influenced the reflex response time of the peroneus longus muscle.

A study by Vaes, Van Gheluwe, and Duquet (2001) indicated that the peroneal muscles exhibited a longer latency time in a group of subjects with chronic ankle instability (58.9 ± 12.0 ms) compared to a control group (47.7 ± 12.1 ms) in response to a sudden unexpected supination movement of 50°. However, another recent study by the same group of authors indicated that there were no differences in the latency of the peroneal muscles in a group of subjects with chronic ankle instability (58.7 ± 12.5 ms) compared to control group (57.3 ± 10.7 ms) in response to a sudden 50° inversion movement (Vaes, Duquet, & Van Gheluwe, 2002).

Of the seven studies outlined above investigating the reflex response time of the peroneal muscles to a sudden unexpected inversion, four support the hypothesis of articular deafferentation, while the remaining three do not. Thus, there appears to be little agreement in the literature as to whether there is really a delayed reflex peroneal response time to sudden unexpected inversion stress. The reflex response time of the peroneus longus to a sudden inversion perturbation has been shown to be a reliable measure (Benesch, Pütz, Rosenbaum, & Becker, 2000). Difference in the results obtained in the studies outlined above may be the result of the non-homogenous criteria used to define functional instability of the ankle joint. It has recently been suggested that efforts should now be made to standardize subject classification in studies on ankle instability, as unless this is done it will be difficult to interpret research findings (Caulfield, 2006). Furthermore it has been

suggested that muscle spindles may be the initiating sensory organs involved in the modulation of reflex response to joint perturbation (Hogervorst & Brand, 1998), and that surface EMG may not be sensitive enough to detect differences in levels of muscle activation both before and after a perturbation stimulus as a result of gamma-system modulation over muscle spindle response sensitivity (Riemann, Myers, & Lephart, 2002). Another consideration when interpreting the results of the studies outlined above is that of anticipation and expectation, which have been shown to have an effect on muscle reflex latencies (Horak, Diener, & Nashner, 1989). Only the studies by Ebig et al. (1997) and Vaes et al. (2001, 2002) controlled for vestibular and visual cues, whereby subjects were blindfolded and wore earphones to minimize the influence of the descending brain stem level and higher brain centre level commands, which can modulate alpha and gamma motor neuron pool excitability. Furthermore, it must be noted that all authors have used different angles of perturbation which may lead to differences in the reflex response time of the muscles investigated and renders comparison of the results difficult.

There is evidence to suggest that the theory of articular deafferentation and a subsequent decrease in reflex stabilization of the ankle joint may not be the main physiological mechanism underlying the development of functional instability. Konradsen, Ravn, and Sorensen (1993) investigated the peroneal reflex reaction time to sudden ankle inversion before and after regional block (Bier-block at the level of the malleoli to obtain a total block of afferent information from the ankle and foot) of the ankle and foot with local anaesthetic. Anaesthesia of the ankle totally blocked the afferent input from mechanoreceptors in the ligaments and capsule of the ankle. The peroneal reaction time to sudden ankle inversion was not altered (80 ms before and 83 ms under anaesthesia). The authors concluded that information for the response to the sudden inversion was mediated by receptors in the muscle/tendon system surrounding the ankle joint.

The ability of the peroneal musculature to provide dynamic ankle protection against sudden ankle inversion has been questioned (Ashton-Miller, Ottaviani, Hutchinson, & Wojtys, 1996; Konradsen, Voigt, & Hojsgaard, 1997). After the delay due to neural transmission, which typically range from 85 to 90 ms until myoelectric activity is first observed, there is an additional delay because muscle contractile mechanics dictate that a further 90 ms is required by a muscle to develop contractile force to even half-maximal levels (Ottaviani, Ashton-Miller, Kothari, & Wojtys, 1995). It is interesting to compare these times to those required to cause an inversion sprain of the lateral ligament complex of the ankle. Konradsen et al. (1997) have shown that unlimited subtalar inversion from a standing position

would put the lateral ligament complex at risk of sprain after approximately 100 ms. Thus Konradsen et al. (1997) concluded that the ankle musculature cannot react fast enough to protect an ankle from injury in the case of sudden unexpected inversion stress. It therefore seems that functional instability does not result simply from a disorder of peroneal reflex control of ankle stability. In the case that ankle inversion occurs in less than 100 ms, it is clear from the two consecutive time delays aforementioned (85 and 90 ms) that the peroneal musculature must be activated prior to the onset of the external forces during foot-strike to provide dynamic ankle stability. This suggests that there must be a higher level of control of ankle muscle function to afford ankle stability during functional activities, a hypothesis confirmed by the studies outlined in the next section, which refute the theory of articular deafferentation and subsequent decreased ankle joint stabilization.

5. Studies refuting the theory of articular deafferentation leading to decreased ankle joint stabilization

Hertel, Guskiewicz, Kahler, and Perrin (1996) examined the effect of induced anaesthesia of the lateral ankle joint on weight-bearing proprioception as measured by postural sway. In this study subjects reported for testing on two separate occasions in a counterbalanced schedule. Postural sway measures were analysed for the non-anaesthetized and anesthetized (8 cc injection of 1% lidocaine into the anterior talofibular ligament and lateral joint capsule) conditions. ANOVA of postural sway scores revealed no main effect for condition, thus indicating that articular deafferentation may not be associated with the development of postural instability following ankle joint injury. The study of Hertel et al. (1996) is further supported by a more recent study by Riemann, Myers, Stone, and Lephart (2004) who examined the effect of an experimentally induced anaesthesia of the anterior talofibular and calcaneofibular ligaments on postural stability during a single leg stance task as well as during a single leg step down landing task. Results failed to demonstrate a difference between the control and experimental conditions. Thus, the authors concluded that articular deafferentation may not be the process by which subjects develop functional instability of the ankle joint following sprain of the lateral ankle joint ligaments, and that other factors such as central motor programming may be more important. A study by Caulfield, Crammond, O'Sullivan, Reynolds, and Ward (2004) showed that subjects with functional instability of the ankle joint did not exhibit any changes in immediate post ground contact peroneus longus EMG activity, indicating that subjects with functional instability do not experience any changes in reflex activation of the peroneus longus

muscle. This study, however, did show that prior to ground contact during a jump landing technique that subjects with functional instability exhibit a decrease in peroneus longus EMG activity. The authors concluded that subjects with functional instability exhibit changes in motor control thus providing direct evidence of central motor programming changes in subjects with functional instability, and thus agreeing with the hypothesis of [Riemann et al. \(2004\)](#), that changes in central motor programming may be more important than articular deafferentation, and the reflex response of the peroneal muscles in the development of functional instability of the ankle joint.

6. Altered neuromuscular control

One possible explanation for the neuromuscular changes noted in subjects with functional instability of the ankle joint could be the presence of arthrogenic muscle inhibition. Arthrogenic muscle inhibition has been described by [Hopkins and Ingersoll \(2000\)](#) as an ongoing reflex reaction of musculature surrounding a joint after distension or damage to the structures of that joint. [Myers, Riemann, Hwang, Fu, and Lephart \(2003\)](#) investigated the effect of a lidocaine or saline injection into both the anterior talofibular and calcaneofibular ligaments, on ankle joint muscle reflex latencies and protective muscle response following inversion perturbation as well as mean ankle joint muscle amplitude during walking. Results of the study showed that the protective response of the peroneal muscles during inversion perturbation and the mean muscle activation amplitude during running decreased following both injections. The authors concluded that the edema induced by the injection of both solutions may have altered the sensorimotor influence of the lateral ankle joint ligaments thus inhibiting the dynamic stabilizers of the ankle joint. A more recent study by [Hopkins and Palmieri \(2004\)](#) also showed that an experimentally induced ankle joint effusion has an adverse effect on peroneus longus muscle activation. Further support for the functional significance of arthrogenic muscle inhibition as a mediating factor in the development of residual symptoms following lateral ankle sprain comes from a recent study by [McVey, Palmieri, Docherty, Zinder, and Ingersoll \(2005\)](#) who demonstrated that arthrogenic muscle inhibition as evidenced by a decreased H:M ratio (H-reflex: M-wave) is present in subjects with functional instability of the ankle joint.

Arthrogenic muscle inhibition may not only have an effect on the ankle joint evertors but also could affect the ankle joint invertors through the process of selective inhibition. It has shown that invertor strength deficits are present in subjects with functional ankle instability ([Ryan, 1994](#); [Wilkerson, Pinerola, & Caturano, 1997](#); [Munn, Beard, Refshauge, & Lee, 2003](#)), thus suggesting

that invertor strength deficits may play a significant role in the development of residual symptoms following lateral ligament complex ankle sprains. The process of selective inhibition was described by [Swearingen and Dehne \(1964\)](#) who postulated that decreased stress tolerance of an injured joint triggers reflexive mechanisms which inhibit muscles that are capable of increasing tensile stress on damaged ligaments. Thus, the invertors of the ankle may be inhibited due to their ability to initiate movement in the direction of initial injury.

[Wilkerson et al. \(1997\)](#) proposed that the invertor muscles may play a significant role in preventing loss of postural stability over a fixed foot. When the centre of mass is displaced over a fixed foot with both its medial and lateral borders anchored, the shank moves laterally resulting in closed chain eversion. When the centre of mass is displaced beyond the lateral border of the foot and the limit of closed chain eversion is reached, the medial border of the foot will begin to rise, subsequently resulting in a rapid inversion of the foot. Hence, eccentric activity of the invertor muscles, which control lateral postural stability, may play a significant role in the maintenance of dynamic ankle stability. Therefore, if the invertors are weak there may be a predisposition to inversion sprains. This has yet to be established, however, and the correlation of arthrogenic muscle inhibition and decreased invertor muscle strength in subjects with functional instability warrants further investigation.

7. Clinical application

There is evidence to suggest that subjects with functional instability exhibit changes in feed-forward control, whereby they have decreased levels of peroneus longus muscle activation during dynamic activities ([Caulfield et al., 2004](#); [Delahunt, Monaghan, & Caulfield, 2006](#)). Rehabilitation strategies which produce sudden unexpected changes in joint movement should be included to facilitate unconscious joint stabilization. Activities such as training on unstable surfaces as well as training the correct execution of landing techniques could enhance pre-programmed feed-forward synergistic muscular activity. It has been suggested that intrinsically stiffer muscles can facilitate the capabilities of the extrinsic component ([Riemann & Lephart, 2002](#)). Intrinsically stiffer muscles resulting from increased feed-forward muscle activity are thought to enhance muscle spindle activation, resulting in a decreased time to initiation of a reflex response ([Grabner, 1986](#)).

8. Conclusion

The purpose of this manuscript was to provide a comprehensive review of the nature of articular

deafferentation and reflex peroneal muscle stabilization of the ankle joint and their potential role in the development of functional instability of the ankle joint. There is conflicting evidence regarding the extent to which ligament mechanoreceptors induce reflex stabilization of the ankle joint in response to rapid inversion movements. Even if such reflex mechanisms do exist it is unlikely that they would be fast enough to provide adequate joint protection during dynamic activity, thus suggesting that their actual functional relevance is limited.

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Ethical Approval

As this is a review article no ethical permission was necessary from the University ethical review board.

Conflict of interest statement

There were no conflicts of interest in the preparation of this manuscript.

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